



Batch mode testing

Initial Setup

1. Cleanup
2. Execute Run Notebook to complete Setup and History Loader
3. Validate History load

Batch 1

4. Produce Payload 1 to the landing zone
5. Execute Run Notebook to process Payload 1
6. Validate output for Payload 1

Batch 2

7. Produce Payload 2 to the landing zone
8. Execute Run Notebook to process Payload 2
9. Validate output for Payload 2



Stream mode testing

Initial Setup

1. Cleanup
2. Create and trigger a Job to execute Run Notebook
3. Validate History load

Batch 1

4. Produce Payload 1 to the landing zone
5. Validate output for Payload 1

Batch 2

6. Produce Payload 2 to the landing zone
7. Validate output for Payload 2

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📄 Cmd 1

```
1 dbutils.widgets.text("Environment", "dev", "Set the current Environment/catalog name")
2 dbutils.widgets.text("Host", "", "Databricks Workspace URL")
3 dbutils.widgets.text("AccessToken", "", "Secure Access Token")
```

📄 Cmd 2

```
1 env = dbutils.widgets.get("Environment")
2 host = dbutils.widgets.get("Host")
3 token = dbutils.widgets.get("AccessToken")
```

📄 Cmd 3

```
1 %run ../02-setup
```

Python ▶ ▾ - ✕

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1 %run ../02-setup

Cmd 4

```
1 SH = SetupHelper(env)
2 SH.cleanup()
```

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```
1 job_payload = \
2
3     "name": "stream-test",
4     "webhook_notifications": {},
5     "timeout_seconds": 0,
6     "max_concurrent_runs": 1,
7     "tasks": [
8         {
9             "task_key": "stream-test-task",
10            "run_if": "ALL_SUCCESS",
11            "notebook_task": {
12                "notebook_path": "/Repos/SBIT/SBIT/07-run",
13                "source": "WORKSPACE"
14            },
15            "job_cluster_key": "Job_cluster",
16            "timeout_seconds": 0,
17            "email_notifications": {}
18        }
19    ],
20    "job_clusters": [
21        {
22            "job_cluster_key": "Job_cluster",
23            "new_cluster": {
24                "spark_version": "13.3.x-scala2.12",
```

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```
1 #Creating a streaming job
2 import requests
3 import json
4 create_response = requests.post(host + '/api/2.1/jobs/create', data = json.dumps(job_payload), auth = ("token", token))
5 print(f"Response: {create_response}")
6 job_id = json.loads(create_response.content.decode('utf-8'))["job_id"]
7 print(f"Created Job {job_id}")
```

Cmd 7

```
1 #Triggering the streaming job
2 run_payload = {"job_id": job_id, "notebook_params": {"Environment": env, "RunType": "stream", "ProcessingTime": "1
seconds"}}
3 run_response = requests.post(host + '/api/2.1/jobs/run-now', data = json.dumps(run_payload), auth = ("token", token))
4 run_id = json.loads(run_response.content.decode('utf-8'))["run_id"]
5 print(f"Started job id {run_id}")
```

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```
4 run_id = json.loads(run_response.content.decode('utf-8'))["run_id"]
5 print(f"Started job id {run_id}")
```

Cmd 8

```
1 #Wait until job starts
2 import time
3 status_payload = {"run_id": run_id}
4 job_status = "PENDING"
5 while job_status == "PENDING":
6     time.sleep(20)
7     status_job_response = requests.get(host + '/api/2.1/jobs/runs/get', data = json.dumps(status_payload), auth =
8     ("token", token))
9     job_status = json.loads(status_job_response.content.decode('utf-8'))["tasks"][0]["state"]["life_cycle_state"]
10    print(job_status)
```

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1 %run ./03-history-loader

Cmd 10

1 %run ./10-producer

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1 %run ./04-bronze

Cmd 12

1 %run ./05-silver

Python

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Run all

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1 %run ./06-gold

Cmd 14

```
1 import time
2
3 print("Sleep for 2 minutes and let the setup and history loader finish...")
4 time.sleep(2*60)
5
6 #Validate setup and history load
7 HL = HistoryLoader(env)
8 PR = Producer()
9 BZ = Bronze(env)
10 SL = Silver(env)
11 GL = Gold(env)
12
13 SH.validate()
14 HL.validate()
```

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```
17 PR.produce(1)
18 PR.validate(1)
```

Cmd 15

```
1 print("Sleep for 2 minutes and let microbatch pickup the data...")
2 time.sleep(2*60)
3
4 #Validate gold, silver and bronze layer
5 BZ.validate(1)
6 SL.validate(1)
7 GL.validate(1)
8
9 #Produce some incremental data and wait for micro-batch
10 PR.produce(2)
11 PR.validate(2)
```

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10 PR.produce(2)
11 PR.validate(2)

Cmd 16

Python

```
1 print("Sleep for 2 minutes and let microbatch pickup the data...")
2 time.sleep(2*60)
3
4 #Validate bronze, silver and gold layer
5 BZ.validate(2)
6 SL.validate(2)
7 GL.validate(2)
```

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Share

```
6 SL.validate(2)
7 GL.validate(2)
```

Cmd 17

```
1 #Terminate the streaming job
2 cancel_payload = {"run_id": run_id}
3 cancel_response = requests.post(host + '/api/2.1/jobs/run/cancel', data = json.dumps(cancel_payload), auth = ("token",
4 print(f"Canceled job run {run_id}. Status {cancel_response}")
```

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Share

```
token))
4 print(f"Canceled job run {run_id}. Status {cancel_response}")
```

Cmd 28

```
1 #Delete the job
2 delete_job_payload = {"job_id": job_id}
3 delete_job_response = requests.post(host + '/api/2.1/jobs/delete', data = json.dumps(delete_job_payload), auth =
  ("token", token))
4 print(f"Canceled job run {run_id}. Status {delete_job_response}")
```

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